Reproducible Comparison and Validation of Low-Earth-Orbit Propagation Models

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Abstract

This paper compares five low-Earth-orbit (LEO) propagation configurations: analytical two-body propagation, numerical two-body propagation, numerical propagation with the second zonal harmonic J2, numerical propagation with J2 and a simplified atmospheric-drag model, and Simplified General Perturbations 4 (SGP4) propagation from a fixed two-line element set (TLE). The Python workflow was evaluated using two controlled LEO cases and one International Space Station (ISS) case. All 11 production experiments and 36 convergence cases completed successfully. Over seven days, the analytical and numerical two-body solutions differed by no more than 0.000253 m, confirming negligible numerical integration error for the selected solver settings. Comparing the J2 and two-body solutions produced maximum three-dimensional position differences of 5,055.983 km for the 400 km case and 1,487.420 km for the 700 km case. The fitted right-ascension-of-the-ascending-node (RAAN) rates agreed with first-order J2 theory within 0.257% and 0.180%. In the 72 h fixed-TLE comparison, the numerical J2-plus-drag model remained within 10.717 km of SGP4. For the same case, the maximum J2-plus-drag versus SGP4 differences in geometric acquisition and loss times were 1.685 s and 1.127 s. Independent General Mission Analysis Tool (GMAT) comparisons, archived configurations, and file checksums support traceability and reproducibility. The study compares propagation-model behaviour under controlled assumptions. It does not assess true-orbit accuracy or high-fidelity atmospheric prediction.

Keywords: orbit propagation, low Earth orbit, J2 perturbation, atmospheric drag, SGP4, GMAT, reproducibility

Nomenclature

A = spacecraft reference area, m²
a = semimajor axis, km
C_D = drag coefficient
e = eccentricity
H = atmospheric scale height, km
i = inclination, rad
J_2 = second zonal harmonic coefficient
m = spacecraft mass, kg
n = mean motion, rad/s
p = semilatus rectum, km
R_E = Earth reference radius, km
r = magnitude of the position vector, km
r = inertial position vector, km
v = inertial velocity vector, km/s
v_{rel} = velocity relative to the rotating atmosphere, km/s
Δr = position difference, km

Δv = velocity difference, km/s
μ = Earth gravitational parameter, km^3/s^2
ρ = atmospheric density, kg/m^3
Ω = right ascension of the ascending node, rad

I. Introduction

Orbit propagation supports mission design, ground-station planning, satellite operations, and many other spaceflight activities. The required propagation model depends on the purpose of the analysis. A simple model offers speed, transparency, and low computational cost. A more detailed model becomes necessary when longer propagation periods, orbital perturbations, or small timing differences affect the result.

The two-body model treats Earth as a point mass and provides a useful reference because its assumptions are clear and its analytical solution is well established. A numerical two-body solution also provides a direct check of the numerical integrator. The second zonal harmonic, J2, represents the dominant effect of Earth's oblateness for many low-Earth-orbit (LEO) applications and produces long-term changes in the orbital plane. Atmospheric drag becomes important at lower altitudes, but realistic drag prediction requires atmospheric-density data, space-weather inputs, spacecraft attitude, and a well-defined ballistic coefficient. Simplified General Perturbations 4 (SGP4) serves a different purpose. SGP4 propagates mean orbital elements supplied through a two-line element set (TLE) and should not be treated as the same physical model as a Cartesian numerical propagator [1,10].

Propagation comparisons are easy to misinterpret. Differences between two predictions may result from the force model, initial state, reference frame, Earth constants, numerical settings, output times, or orbital-element convention. Earth orientation and polar motion also influence high-precision comparisons between independent software tools [2,9,12]. A valid comparison must therefore control these inputs before differences are attributed to the propagation models.

The present study addresses this problem through a controlled and reproducible comparison of analytical two-body propagation, numerical two-body propagation, numerical propagation with J2, numerical propagation with J2 and simplified atmospheric drag, and fixed-TLE SGP4 propagation. Two controlled LEO cases at approximately 400 km and 700 km altitude are used to examine force-model effects. An International Space Station (ISS) case is used to compare the numerical models with SGP4 and to evaluate geometric ground-station pass timing. Common initial conditions, fixed configuration files, convergence testing, independent GMAT comparisons, and archived SHA-256 checksums are used to support traceability.

The main research question is:

How do model choice and propagation duration affect trajectory separation, nodal motion, geometric ground-station pass timing, and computational cost for representative low-Earth-orbit cases?

The paper makes four practical contributions. First, analytical and numerical two-body propagation are compared to separate numerical integration error from force-model differences. Second, the effects of J2 and simplified atmospheric drag are measured for the controlled 400 km and 700 km cases. Third, the numerical models are compared with a fixed-TLE SGP4 history for an International Space Station case, including position separation and geometric pass-boundary timing. Fourth, every reported result is linked to saved configurations, test records, GMAT evidence, figures, tables, and file checksums.

The study focuses on model behaviour under controlled assumptions. The reported differences do not represent true-orbit errors because no precise tracking solution is used as an observational reference. The simplified drag model also serves as a sensitivity model rather than a high-fidelity atmospheric prediction method.

II. Models and Methods

A. Software and reproducibility workflow

The propagation workflow was implemented in Python. The numerical models used SciPy's DOP853 integrator. The fixed-TLE case used the Python sgp4 implementation, and Astropy supplied the registered coordinate transformations [4,9,10]. Each experiment was controlled through saved configuration files containing the orbit case, initial epoch, propagation duration, force model, numerical settings, physical constants, and output interval.

The production campaign contained three orbital test cases, 11 case-duration combinations, and 43 primary model runs. CASE-LEO400 used four configurations at each of four durations. CASE-SSO700 used three primary configurations at each of four durations; four J2-plus-drag outputs were retained as secondary sensitivity runs. The fixed-TLE ISS case used five configurations at each of three durations. The complete campaign therefore executed 47 model runs.

Before the production runs, a 24 h convergence study evaluated 36 DOP853 solver configurations. The matrix combined three relative tolerances (10^{-7}, 10^{-9}, and 10^{-11}), three absolute tolerances (10^{-9}, 10^{-11}, and 10^{-13}), and four maximum internal step sizes (120, 60, 30, and 10 s). Every configuration was compared with the analytical two-body solution and a stricter numerical reference. Energy conservation, angular-momentum conservation, runtime, and force-function evaluations were also recorded. The final production runs used a relative tolerance of 10^{-11}, an absolute tolerance of 10^{-13}, and a maximum internal step of 60 s.

The recorded production environment was CPython 3.14.5, SciPy 1.18.0, sgp4 2.27, Astropy 7.2.2, NumPy 2.5.1, and Matplotlib 3.11.1 on Windows 11 build 26200. The host reported eight logical processors and an AMD64 Family 23 Model 24 Stepping 1 processor. These details are stored with each production run.

B. State definition and reference conventions

The dynamic state is expressed in an Earth-centred inertial (ECI) frame and contains Cartesian position ($r(t)$) and velocity ($v(t)$).

$$x(t)=\begin{bmatrix} r(t) \\ v(t) \end{bmatrix} \tag{1}$$

For the controlled 400 km and 700 km cases, classical orbital elements were converted once to a Cartesian state. The analytical and numerical models then started from the same state. This avoids differences caused by separate element-to-state conversions.

Distances were stored in kilometres, velocities in kilometres per second, and time in seconds. The controlled numerical cases used $\mu=398600.4418$ km³/s², $R_E=6378.1363$ km, $J_2=0.00108262668$, and an Earth rotation rate of 7.292115×10^{-5} rad/s. The fixed ISS case used an Earth radius of 6378.137 km. These values were kept fixed throughout the experiments. The frozen benchmark intentionally combines the IERS 2010 gravitational parameter and rotation rate with the EGM96-compatible equatorial radius and J2 coefficient; the ISS geodetic conversion uses the WGS 84 equatorial radius [2,7,8].

The controlled cases used the documented ECI_CONTROLLED benchmark frame, with its origin at Earth's centre and its z axis aligned with the assumed mean rotation pole. This benchmark frame is not a complete standards-based celestial realization. For the ISS case, SGP4 states in the True Equator, Mean Equinox (TEME) frame were transformed by Astropy to the Geocentric Celestial Reference System (GCRS) before comparison. Ground tracks and Bremen access were calculated after transformation from GCRS to the International Terrestrial Reference System (ITRS), using WGS 84 geodetic coordinates and the registered earth-orientation-parameter data [2,9].

C. Analytical and numerical two-body models

The two-body model treats Earth as a point mass. It excludes Earth’s oblateness, atmospheric drag, third-body forces, solar-radiation pressure, relativity, and satellite manoeuvres. The acceleration is

$$\ddot{r} = -\frac{\mu}{r^3} r \,, r = \|r\| \tag{2}$$

The analytical model advances the Keplerian orbit and reconstructs the Cartesian state at each requested output time. The numerical model integrates Eq. (2) directly in Cartesian coordinates. The analytical and numerical solutions use the same central-gravity model. Their difference therefore measures numerical integration and implementation error rather than force-model differences.

The numerical integrations used SciPy’s DOP853 solver, an explicit eighth-order Runge-Kutta method [4]. The selected solver settings were established through the 36-configuration convergence study described in Section II.A.

D. J2 perturbation model

The J_2 model adds the leading zonal-harmonic correction caused by Earth’s equatorial bulge. For a symmetry axis aligned with the inertial z direction, the perturbing acceleration is

$$a_{J_2} = \frac{3 J_2 \mu R_E^2}{2 r^5} \begin{bmatrix} x\left(\frac{5 z^2}{r^2} - 1\right) \\ y\left(\frac{5 z^2}{r^2} - 1\right) \\ z\left(\frac{5 z^2}{r^2} - 3\right) \end{bmatrix} \tag{3}$$

The validation programme also tested a pole-aware form of the zonal acceleration. That form evaluates the perturbation relative to the selected Earth pole and rotates the result consistently. The production runs use the validated convention selected during the GMAT comparison.

The fitted nodal rate was compared with the standard first-order expression [5].

$$\dot{\Omega} = -\frac{3}{2} J_2 n \left(\frac{R_E}{p}\right)^2 \cos i \,,\; p = a\left(1 - e^2\right) \tag{4}$$

where (n) is the mean motion, (a) is the semimajor axis, (e) is the eccentricity, (i) is the inclination, and (p) is the semilatus rectum.

The J2 effect was evaluated in two separate ways.

First, the numerical J2 trajectory was compared with the numerical two-body trajectory. This comparison measured the three-dimensional trajectory separation caused by adding the J2 perturbation.

Second, a linear trend was fitted to the right ascension of the ascending node (RAAN) history from the numerical J2 simulation. The resulting average nodal rate was compared with the theoretical value from Eq. (4).

The trajectory-separation comparison and the nodal-rate comparison answer different questions. The first measures how far the predicted positions move apart. The second checks whether the simulated orbital-plane rotation agrees with first-order J2 theory.

Equation (4) provides a secular trend estimate. It is not expected to reproduce every short-period variation in the numerical RAAN history.

E. Simplified drag model

The drag model was included as a sensitivity model, not as an operational atmosphere model. The acceleration is

$$a_D = -\frac{1}{2} \rho C_D \frac{A}{m} \left\| v_{rel} \right\| v_{rel} \tag{5}$$

where (ρ) is atmospheric density, (C_D) is the drag coefficient, (A) is the reference area, (m) is spacecraft mass, and (v_{rel}) is the spacecraft velocity relative to the rotating atmosphere. The density follows a single-scale-height exponential profile, Equation (5) was evaluated in SI units: the stored relative velocity was converted from km/s to m/s before the drag calculation, and the resulting acceleration was converted from m/s² to km/s² before numerical integration.

$$\rho(h) = \rho_{ref} \exp\left[-\frac{h - h_{ref}}{H}\right] \tag{6}$$

where (h) is altitude, (h_{ref}) is the reference altitude, (ρ_{ref}) is the reference density, and (H) is the scale height. For the controlled 400 km and 700 km cases, the fixed values were $m = 500$ kg, $A = 4$ m², $C_D = 2.2$, $h_{ref} = 400$ km, $\rho_{ref} = 3 \times 10^{-12}$ kg/m³, and $H = 60$ km. For the ISS comparison, the illustrative values were $m = 420000$ kg and $A = 400$ m², while the same C_D, reference density, and scale height were used. The atmosphere was assumed to rotate with Earth.

These parameters were not fitted to tracking data or to the TLE B^* term. The results therefore show how the selected model responds to drag. They do not represent high-fidelity atmospheric-density or operational orbit-decay.

F. Fixed-TLE SGP4 comparison

The fixed ISS case used NORAD catalogue number 25544 and a TLE epoch of 2026-07-18 02:08:01.938048 UTC. The archived TLE was retrieved from CelesTrak and propagated with SGP4 for 6, 24, and 72 h [11]. The same TLE and epoch were retained throughout each comparison; the element set was not updated during propagation.

The raw SGP4 TEME state at the TLE epoch was transformed to GCRS with Astropy [9]. That transformed Cartesian state initialized the analytical two-body, numerical two-body, numerical J2, and numerical J2-plus-drag models. The SGP4 history was treated as a TLE-compatible model reference rather than as measured orbital truth.

SGP4 propagates mean elements supplied through a TLE. The numerical propagators integrate Cartesian position and velocity using selected force equations. Agreement or disagreement between these approaches therefore reflects differences in model formulation, force representation, and element conventions.

The comparison recorded maximum position separation, final position separation, geometric pass count, matched pass count, and differences in acquisition of signal (AOS) and loss of signal (LOS) times.

G. Difference measures

The histories were compared at common output epochs. Position and velocity differences were calculated as

$$\begin{aligned} \Delta r(t) &= \left\| r_1(t) - r_2(t) \right\|, \\ \Delta v(t) &= \left\| v_1(t) - v_2(t) \right\|. \end{aligned} \tag{7}$$

The analysis reports the maximum difference during the propagation interval and the difference at the final epoch. Selected comparisons also use radial, transverse, and normal components. The radial component acts along the position direction. The transverse component acts in the direction of orbital motion within the orbital plane. The normal component acts perpendicular to the orbital plane. Changes in semimajor axis and specific orbital energy are used to describe drag sensitivity.

H. Ground-station access and runtime

The pass analysis used a Bremen station at 53.0793 degrees N, 8.8017 degrees E, and 12 m altitude, with a 10 degree minimum-elevation mask. A 20 s coarse grid located candidate crossings. Piecewise cubic Hermite interpolation and Brent root finding refined AOS and LOS, and passes were matched one-to-one by maximum-elevation time within a 600 s window. Terrain, buildings, refraction, antenna patterns, Doppler, link margin, scheduling, and station downtime were not included. The reported times therefore represent geometric access boundaries rather than complete communications-link availability.

Runtime was measured with wall-clock time on the production computer. These measurements compare models within one implementation and one hardware environment. They should not be interpreted as universal software-performance benchmarks.

I. Verification and GMAT validation

Verification examined whether the equations, numerical methods, and software implementation behaved as intended. The programme included analytical-to-numerical two-body comparison, energy and angular-momentum conservation checks, automated unit tests, the 36-configuration convergence study, repeatable artifact generation, and registered file checksums.

External validation used General Mission Analysis Tool (GMAT) R2026a [3]. Ten multicase two-body and J2 comparisons passed with scope warnings across registered altitudes, inclinations, durations, and earth-orientation settings. A separate earth-orientation series passed 6 of 6 cases and established the full earth-orientation-parameter configuration as the independent baseline. The exponential-drag acceleration comparison passed 4 of 4 scenarios and 25 of 25 checks for density, direction, and mass-area-drag-coefficient scaling. Its maximum acceleration-vector difference was 3.213×10^{-17} km/s².

The validation status includes warnings where the model scope or reference-frame convention limits the interpretation. A warning does not mean that a test failed. It records a known difference or a restriction on the claim.

III. Experiment Design

Three orbit cases were used.

CASE-LEO400 and CASE-SSO700 used the frozen osculating classical elements listed in Table 1. Both controlled cases started at 2026-01-01 00:00:00 UTC and were propagated for 6, 24, 72, and 168 h.

Parameter	**CASE-LEO400**	**CASE-SSO700**
Epoch, UTC	2026-01-01 00:00:00 UTC	2026-01-01 00:00:00 UTC
Semimajor axis, km	6778.1363	7078.1363
Eccentricity	0.001	0.001
Inclination, deg	51.6	98.187964
RAAN, deg	20.0	20.0
Argument of perigee, deg	30.0	30.0
True anomaly, deg	0.0	0.0
Frame	ECI_CONTROLLED	ECI_CONTROLLED

Table 1. Frozen initial conditions for the two controlled orbit cases.

The third case used the fixed ISS TLE identified above. Its raw SGP4 TEME state at the exact TLE epoch initialized the shared GCRS comparison state. This case was propagated for 6, 24, and 72 h and supplied the SGP4, ground-track, and Bremen pass-timing comparisons.

Within each comparison, the initial state, epoch, Earth constants, output times, and numerical settings were held fixed. The model hierarchy was:

1. analytical two-body;
2. numerical two-body;
3. numerical two-body with J_2;
4. numerical two-body with J_2 and simplified drag;
5. SGP4 for the fixed-TLE case.

The convergence study tested 36 combinations of tolerance and maximum step size. The production runs used DOP853 with a relative tolerance of 10^{-11}, an absolute tolerance of 10^{-13}, and a 60 s maximum internal step.

IV. Results

A. Numerical verification

All 47 executed model runs across 11 case-duration combinations completed successfully. The reported production set comprised 43 primary runs and four secondary CASE-SSO700 drag-sensitivity runs. All 36 convergence cases met their registered acceptance criteria.

The analytical and numerical two-body histories remained close over seven days. At 168 h, the maximum position difference was 0.000253 m for CASE-LEO400 and 0.000239 m for CASE-SSO700. These differences are much smaller than the changes caused by J_2 or drag. The selected solver settings therefore contributed little to the reported model-to-model separation.

B. Effect of J2

Figure 1 shows the maximum separation between the J_2 and two-body histories. The difference grew with propagation time in both controlled cases. For CASE-LEO400, the maximum separation increased from 190.900 km at 6 h to 5,055.983 km at 168 h. For CASE-SSO700, it increased from 59.128 km to 1,487.420 km.

Figure 2 compares the fitted nodal rates with first-order analytical theory. For CASE-LEO400, the analytical value was -5.002333 deg/day and the fitted numerical value was -5.015202 deg/day, a relative difference of 0.257%. For CASE-SSO700, the corresponding values were 0.985647 deg/day and 0.987420 deg/day, a relative difference of 0.180%. Both rates followed the expected secular direction.

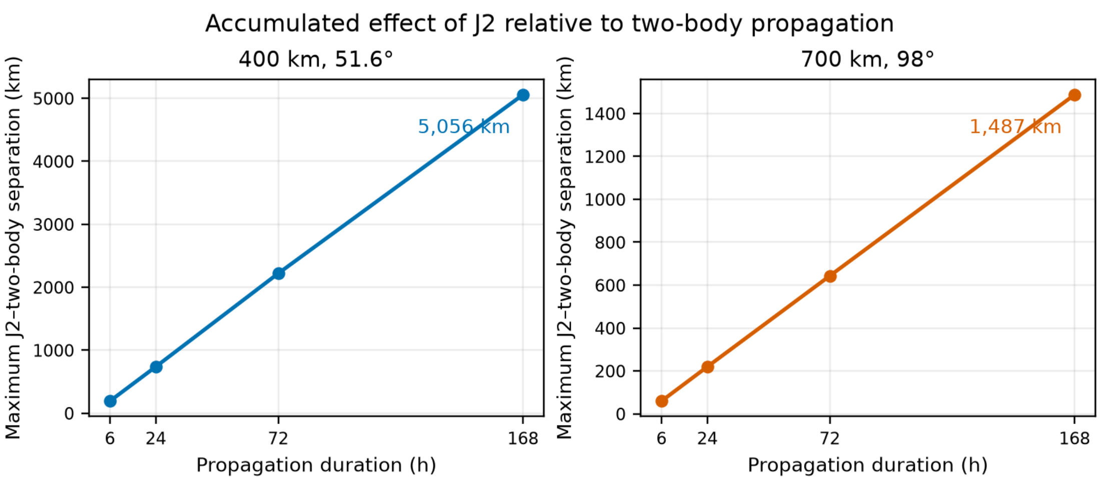

Figure 1. Maximum three-dimensional separation between the numerical J2 and numerical two-body trajectories for the controlled 400 km and 700 km cases.

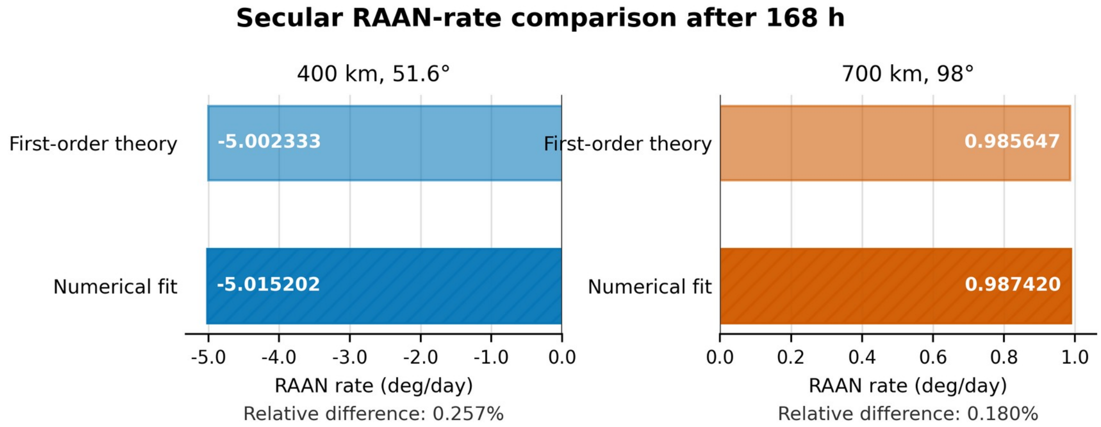

Figure 2. First-order analytical and fitted numerical J2 RAAN rates over the 168 h interval for the controlled 400 km and 700 km cases.

Table 2 summarises the seven-day controlled-case results.

Case	Max. two-body difference, m	Max. J2 separation, km	Theoretical / fitted RAAN rate, deg/day	RAAN difference, %	Max. drag separation, km	Final semimajor-axis change relative to J2, m
CASE-LEO400	0.000253	5,055.983	-5.002333 / -5.015202	0.257	845.759	-3,059.838
CASE-SSO700	0.000239	1,487.420	0.985647 / 0.987420	0.180	5.646	-26.140

Table 2. Controlled-case results at 168 h. The drag columns are relative to the J2 model.

C. Simplified drag sensitivity

Figure 3 shows the separation between the J_2 and J_2 plus drag histories. The 400 km case was far more sensitive to the selected drag model. At 168 h, the maximum separation was 845.759 km for CASE-LEO400 and 5.646 km for CASE-SSO700. The final semimajor-axis changes relative to the J_2 model were -3,059.838 m and -26.140 m.

The maximum separation in the 400 km case was about 149.8 times the value in the 700 km case. This ratio belongs to the selected exponential atmosphere and spacecraft parameters. It should not be treated as a universal altitude relationship. Figure 3 shows the magnitude of the semimajor-axis reduction as a positive quantity.

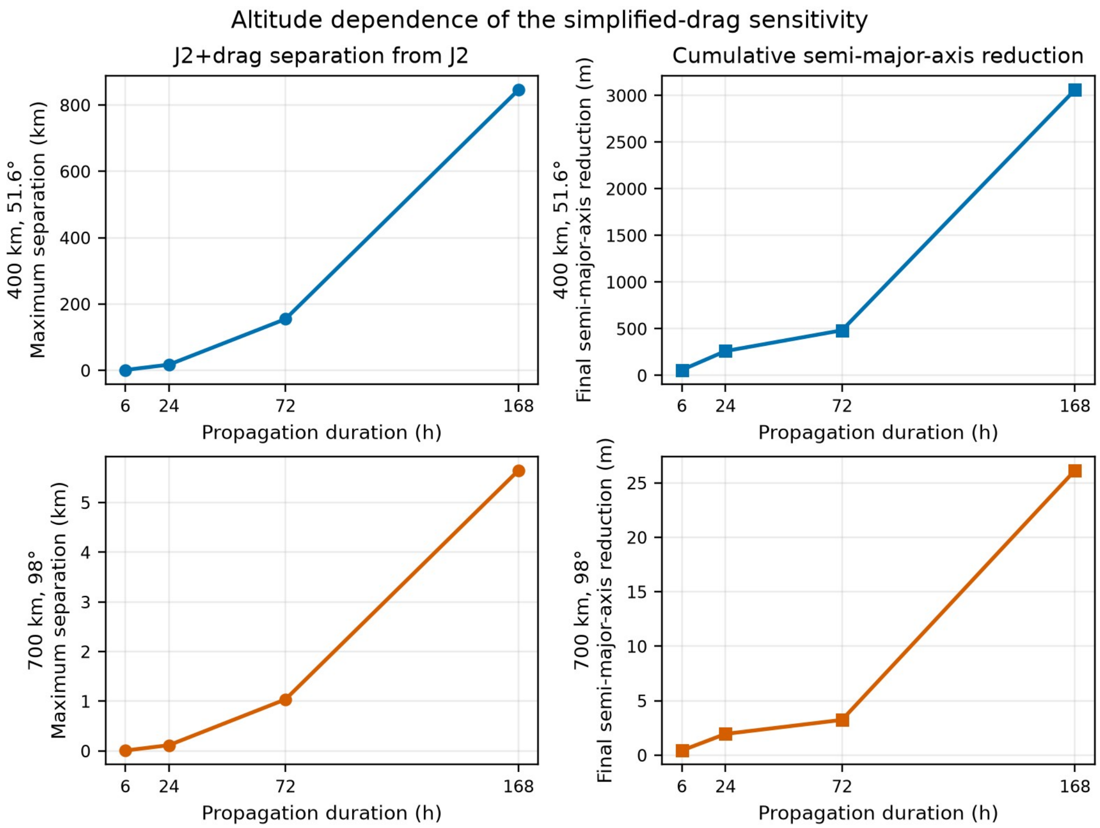

Figure 3. Simplified-drag sensitivity for the controlled 400 km and 700 km cases.

D. Fixed-TLE SGP4 comparison

Figure 4 compares each numerical model with the fixed-TLE SGP4 history. At 72 h, the maximum separation was 3,303.361 km for numerical two-body, 22.736 km for numerical J_2, and 10.717 km for numerical J_2 plus drag. The final separations were 3,303.361 km, 20.680 km, and 8.293 km.

The analytical and numerical two-body curves nearly overlap because both use the same central-gravity model and differ by less than one millimetre over the studied interval. Within this experiment, adding the simplified drag term reduced the 72 h maximum separation from SGP4 by 52.9% relative to the J_2 model. This does not prove higher absolute accuracy. SGP4 and the numerical models use different state and force-model conventions.

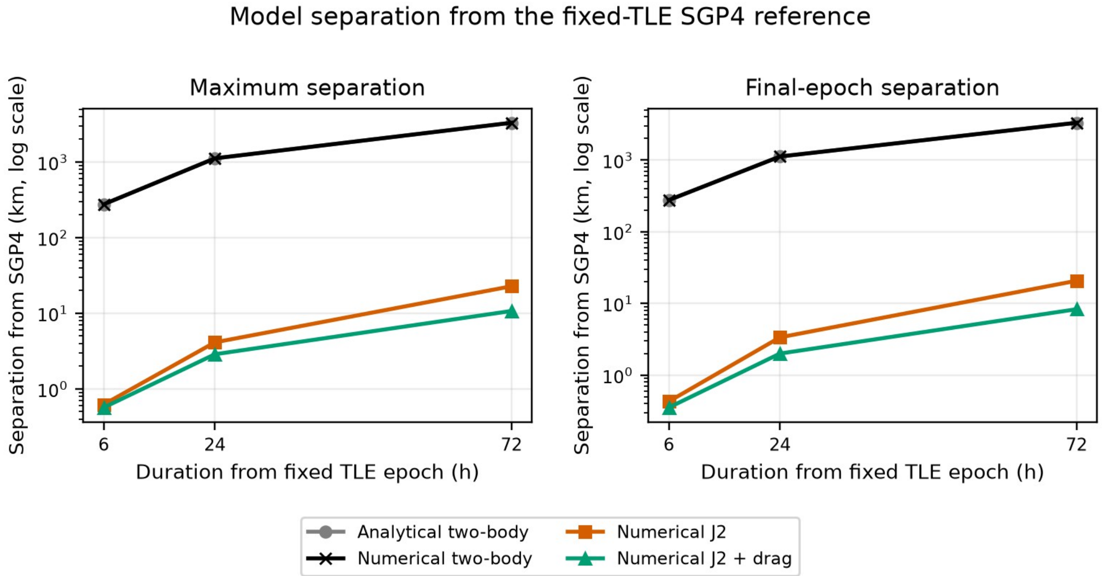

Figure 4. Model separation from the fixed-TLE SGP4 history.

At 24 h, numerical J_2 and numerical J_2 plus drag matched all four reported passes. Their maximum acquisition-time differences were 0.442 s and 0.329 s. Their maximum loss-time differences were 0.327 s and 0.223 s.

At 72 h, both models matched 13 passes. The J_2 plus drag model had maximum acquisition and loss differences of 1.685 s and 1.127 s. Numerical two-body reported 14 passes but matched only 12. Its maximum timing differences exceeded 400 s. The result shows that a large accumulated state difference may change both event timing and the number of detected passes.

Model	Max. separation, km	Final separation, km	Passes	Matched	Max. |AOS difference|, s	Max. |LOS difference|, s
Numerical two-body	3,303.361	3,303.361	14	12	410.547	445.870
Numerical J2	22.736	20.680	13	13	3.143	2.475
Numerical J2 plus drag	10.717	8.293	13	13	1.685	1.127

Table 3. Fixed-TLE SGP4 trajectory separation and geometric pass comparison at 72 h.

E. Runtime

Figure 5 shows the measured runtime for each model. For CASE-LEO400 at 168 h, analytical two-body required 0.755 s, numerical two-body 13.209 s, numerical J_2 27.350 s, and numerical J_2 plus drag 71.813 s. The corresponding times for CASE-SSO700 were 0.702 s, 11.908 s, 24.893 s, and 66.656 s.

For the 72 h ISS case, SGP4 required 0.005611 s. Analytical two-body required 0.295 s, numerical two-body 4.711 s, numerical J_2 9.177 s, and numerical J_2 plus drag 26.522 s.

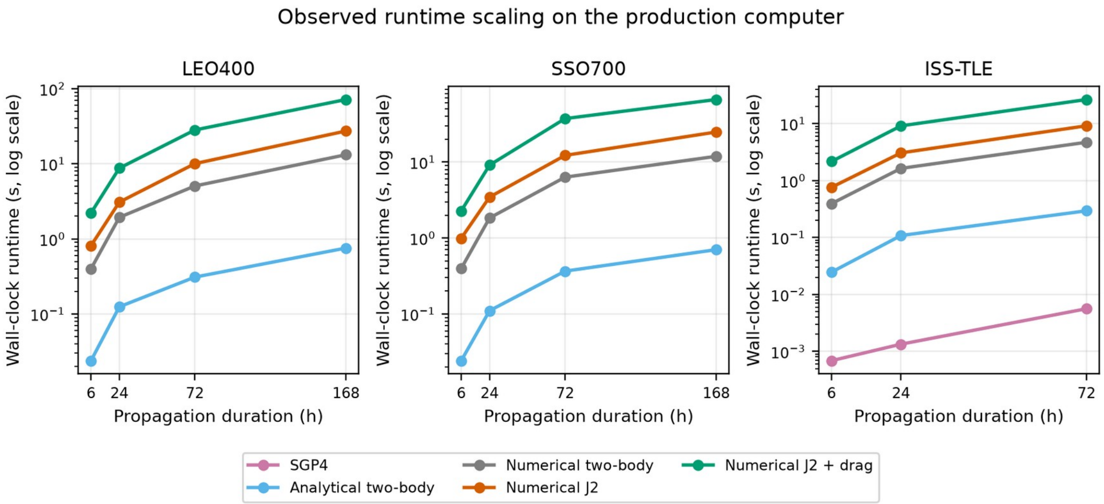

Figure 5. Runtime measured on the production computer.

Case and duration	Model	Runtime, s
CASE-LEO400, 168 h	Analytical two-body	0.755
	Numerical two-body	13.209
	Numerical J2	27.350
	Numerical J2 plus drag	71.813
CASE-SSO700, 168 h	Analytical two-body	0.702
	Numerical two-body	11.908
	Numerical J2	24.893
	Numerical J2 plus drag	66.656
ISS TLE, 72 h	Analytical two-body	0.295
	Numerical two-body	4.711
	Numerical J2	9.177
	Numerical J2 plus drag	26.522
	SGP4	0.005611

Table 4. Selected runtime results. Values apply only to the recorded production computer and software environment.

F. Validation status

The production experiments and convergence tests met their defined gates. The GMAT validation work supported the two-body and J_2 implementations and identified the importance of Earth-orientation conventions. For external validation, GMAT R2026a with the registered full Earth-orientation-parameter dataset was adopted as the reference configuration. This setup reduced differences caused by inconsistent Earth-orientation and frame conventions.

Validation item	Result
Production matrix	11/11 experiments; 43 primary runs; 47 total executions; 0 failed
Convergence study	36/36 DOP853 configurations passed
Analytical versus numerical two-body	Maximum position difference 0.000253 m over 168 h
J2 secular-rate check	Relative differences 0.257% for LEO400 and 0.180% for SSO700
GMAT two-body and J2 multicase matrix	10/10 cases passed with documented scope warnings
GMAT earth-orientation baseline	6/6 independent cases passed; full-EOP configuration adopted
GMAT exponential-drag acceleration	4/4 scenarios and 25/25 checks passed; maximum vector difference 3.213×10^{-17} km/s²

Table 5. Summary of numerical and external validation evidence.

V. Discussion

A. Numerical error and force-model error

The two-body comparison shows that the selected numerical settings were sufficiently accurate for this experiment. The largest analytical-to-numerical position difference remained below 0.000253 m over seven days. In contrast, adding J_2 changed the trajectory by hundreds to thousands of kilometres. For these cases, model choice had a much larger effect than integration error.

The RAAN comparison provides a separate check of the J_2 implementation. The fitted rates differed from first-order theory by less than 0.3%. The analytical expression is approximate and does not include short-period behaviour, so agreement in the secular rate is the relevant result.

B. Meaning of the drag results

The simplified drag model had a much larger effect at 400 km than at 700 km. This is expected because density decreases strongly with altitude. The numerical values, however, depend on the selected density profile, mass, area, and drag coefficient. A realistic thermospheric prediction would require time-dependent atmosphere and space-weather inputs, a better representation of spacecraft attitude and area, and a calibrated ballistic coefficient [6].

The drag results are therefore useful for sensitivity and software testing. They should not be used as a forecast of real ISS decay or as a general rule for all satellites at the same altitude.

C. Meaning of the SGP4 comparison

The J_2 plus drag model remained closer to SGP4 than the J_2 model in the fixed 72 h experiment. This is a useful observation, but it is not a direct accuracy ranking. SGP4 propagates TLE-compatible mean elements [1]. The numerical models propagate a Cartesian state using a different force model. The measured separation includes differences in initialization, element convention, frame treatment, and perturbation modelling.

The pass results show why these differences matter in practice. The J_2 and J_2 plus drag models retained all 13 matched passes at 72 h, while the two-body model produced a pass-count mismatch and timing errors above 400 s. A model that is adequate for a short state estimate may be inadequate for longer access planning.

D. Cost and model selection

Greater model complexity increased runtime. For CASE-LEO400 at 168 h, the drag calculation increased runtime from 27.350 s for J_2 to 71.813 s for J_2 plus drag. This cost is still small for offline studies. SGP4 was much faster in the fixed-TLE case.

The model should match the task. SGP4 is appropriate for TLE-based tracking and catalogue-style propagation. Two-body propagation is useful for teaching, quick estimates, and numerical verification. A J_2 model is needed when nodal motion and multi-day LEO behaviour matter. Drag should be included only when the atmosphere and spacecraft parameters support the intended claim.

E. Validation and generality

The GMAT comparisons increase confidence that the implemented equations and frame handling are consistent with the selected reference setup [3]. They do not make the simplified physical models complete. Validation means that the software reproduced the chosen model within the stated tolerances and conventions.

The experiment uses two controlled near-circular cases and one fixed ISS TLE. Different epochs, eccentricities, inclinations, solar conditions, spacecraft properties, station locations, or output intervals will produce different numerical values. The workflow is reusable. The reported kilometre and timing values belong to the exact cases described in this paper.

VI. Limitations

The controlled cases use the defined ECI_CONTROLLED benchmark axes rather than a complete standards-based celestial frame. The ISS case uses Astropy TEME-to-GCRS and GCRS-to-ITRS transformations, so the results depend on the registered earth-orientation data and those transformation conventions. Ground-track and access analyses were performed only for the fixed-TLE ISS case. The drag model uses a fixed exponential density law and fixed spacecraft parameters. It excludes space-weather history, atmospheric composition, density assimilation, attitude-dependent area, and a calibrated ballistic coefficient.

The SGP4 comparison uses one fixed TLE and does not use measured orbit data. The ground-station analysis includes geometry and a minimum-elevation mask only. It excludes terrain, buildings, refraction, antenna patterns, Doppler, link margin, scheduling, and station downtime.

Runtime results apply to the recorded computer and software environment. The study does not include manoeuvres, orbit determination, covariance propagation, conjunction assessment, third-body gravity, solar-radiation pressure, relativity, or higher-order gravity harmonics. The frozen Earth constants and their sources are stated in Section II.B [2,7,8].

VII. Conclusions

The experiments show that force-model choice dominates numerical integration error for the selected multi-day LEO cases. Analytical and numerical two-body propagation agreed to well below one millimetre over seven days. Omitting J_2 produced position differences of 5,055.983 km at 400 km altitude and 1,487.420 km at 700 km altitude. The fitted nodal rates agreed with first-order theory within 0.3%.

The simplified drag response was much larger in the 400 km case than in the 700 km case, but the result depends on the chosen atmosphere and spacecraft parameters. In the fixed ISS TLE comparison, the J_2 plus drag model remained closer to SGP4 than the J_2 model and matched geometric pass boundaries within two seconds over 72 h.

These results support practical model selection within the stated scope. They do not establish universal propagator accuracy, true-orbit accuracy, or atmospheric prediction quality.

Data and Software Availability

The verified source code, frozen configurations, figures, tables, and SHA-256 checksum manifests supporting this study are available from the corresponding author. A public GitHub release and archival identifier will be added after the final repository review. No public repository claim is made in this manuscript before that release is accessible.

Artificial-Intelligence Assistance Disclosure

AI-assisted tools supported code generation, document structuring, and language editing. The author defined the research questions and physical scope, executed the Python and GMAT workflows, checked the outputs, resolved validation issues, reviewed every scientific statement, and approved the final manuscript. The author is responsible for all submitted material.

Acknowledgments

The author acknowledges the developers and maintainers of Python, SciPy, SGP4 reference implementations, and the General Mission Analysis Tool.

Funding

This research received no external funding.

Conflict of Interest

The author declares no conflict of interest.

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